GRIDZILLA

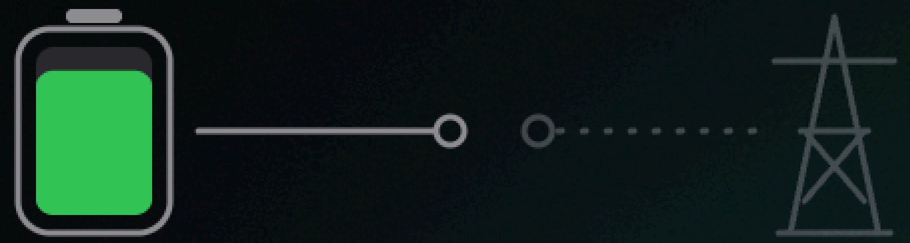
A grid signal for home batteries, cars and heat pumps.

Mathew Stafurik · Peter Mikeška · Matthew · Tedi
Grid-Serving Storage · Energy Bridge Vienna 2026

Europe's largest battery is in people's basements.

The grid can use none of it.

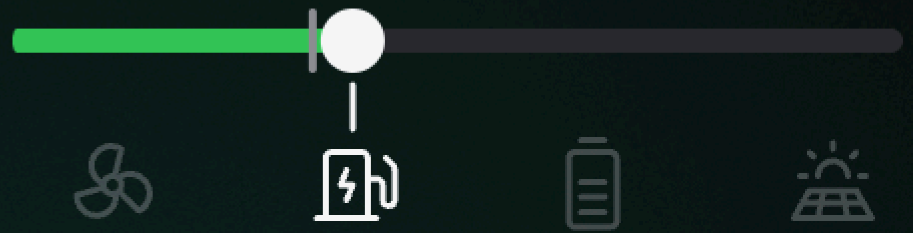
Slovakia



0 signals

from the grid operator
to home batteries

Germany



4.2 kW

the operator may dim
one new charger to this

Ukraine

3 GWh

of home batteries, idle
through the blackouts

The storage exists. The signal does not.

✓ From the grid operator, signed

“This street, next three hours, take at most 3 kW per house.”

Battery ✓

Victron

or GoodWe, any brand

Battery covers the house

Heat pump ✓

Viessmann

Hot water waits until later

Any brand ✓

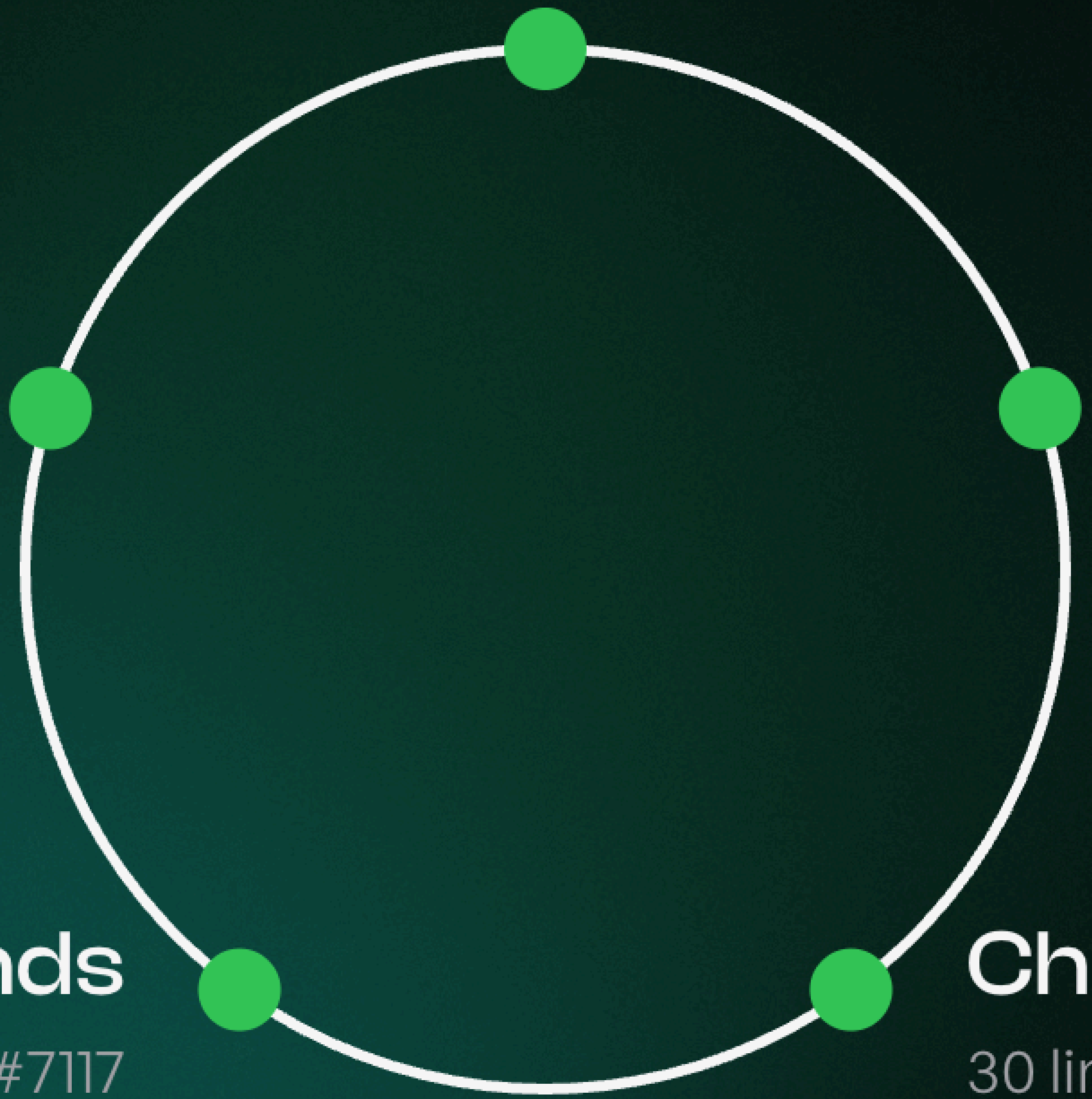
Car charger

Car charges after dinner

Battery first. Car second. Hot water third.

The operator asks. The house decides. **The owner can always say no.**

The loop, every 15 minutes.



Gathers

72 transformers, the weather, the plan, every reply

Thinks

The AI writes the message in 1.5 s

Checks

30 lines of plain rules, not AI

Signs and sends

The platform signs message #7117

Houses reply

271 homes answer, all signed

Six words. Signed.

SUPPORT

Evening peak, or the grid short of power

"Take at most 3 kW from the street."

ABSORB

Noon sun flooding the street

"Send back at most 2 kW."

PREPARE

A warning: storm, planned outage, attack

"Be 90 % full by 17:00."

STRETCH

The street is dark, power back at a known time

"Run on your own, send nothing back, make it last."

RECOVER

The transformer is replaced

"Refill slowly, not all at once."

RELEASE

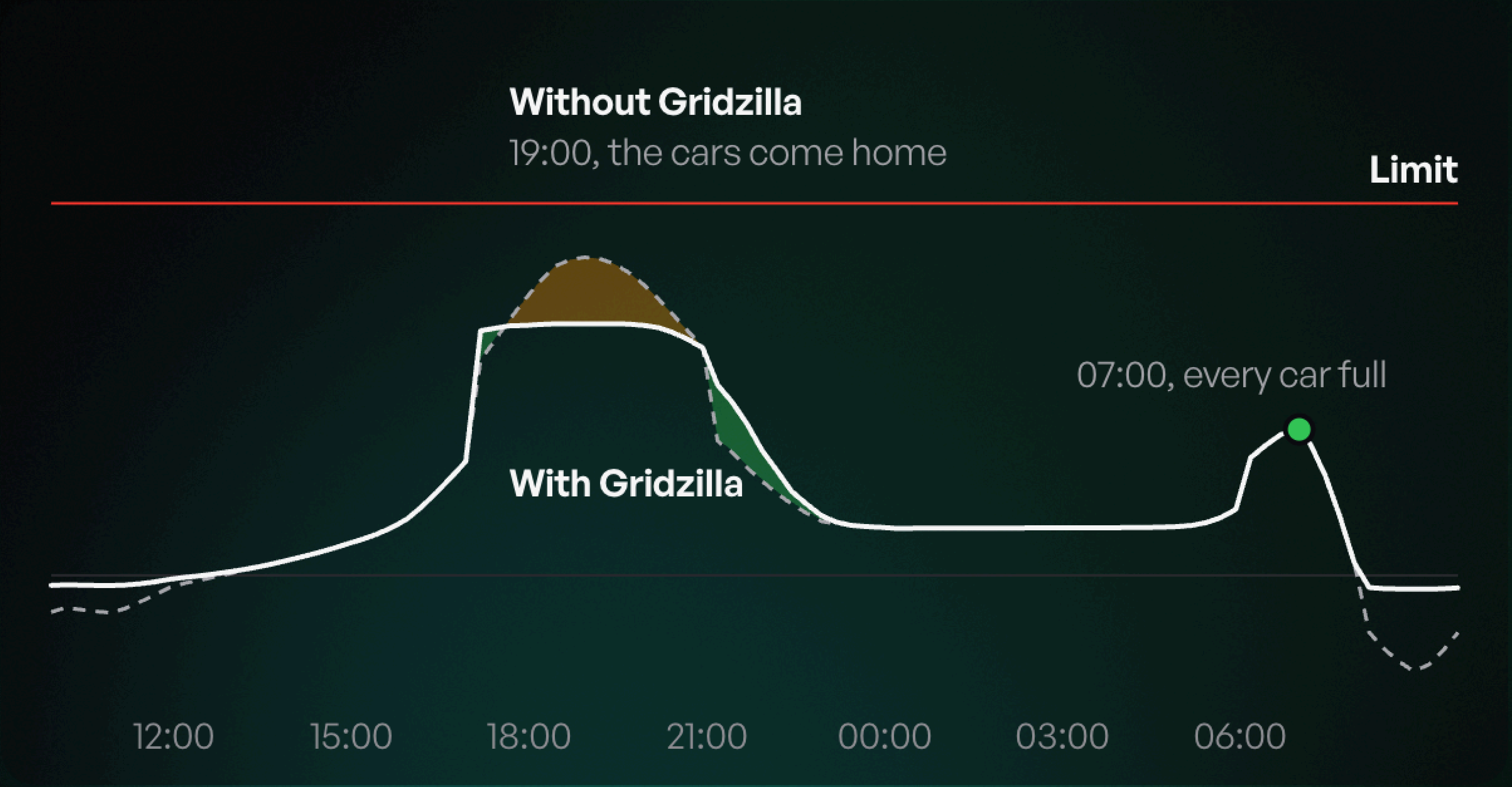
The event is over

"You are on your own again."

Plus a 24-hour envelope per street, renewed every two hours. Every message Ed25519-signed with a time window, ramps randomised so a street never steps in sync, the owner's limits always win, and everything reverts when the heartbeat stops.

The monster eats peaks.

Every street, every evening.



68 %

with Gridzilla, tonight's plan down from 85 % without it

The next 24 hours for Vienna: 2,168 homes, cars and heat pumps, renewed every two hours from the real weather. Live.

Saved tonight

5.3 MW
off the city's evening peak

90 kW
added right now: charging moved into the night

2 of 2
streets kept inside their limit

10,419 kWh
car charging moved into the night

What we did

0
signals signed today

0
houses following a signal now

Now autopilot off · off

Sees
Autopilot off

Thinks
Resting

Checks
Standing by

Acts
Nothing sent

Signals running

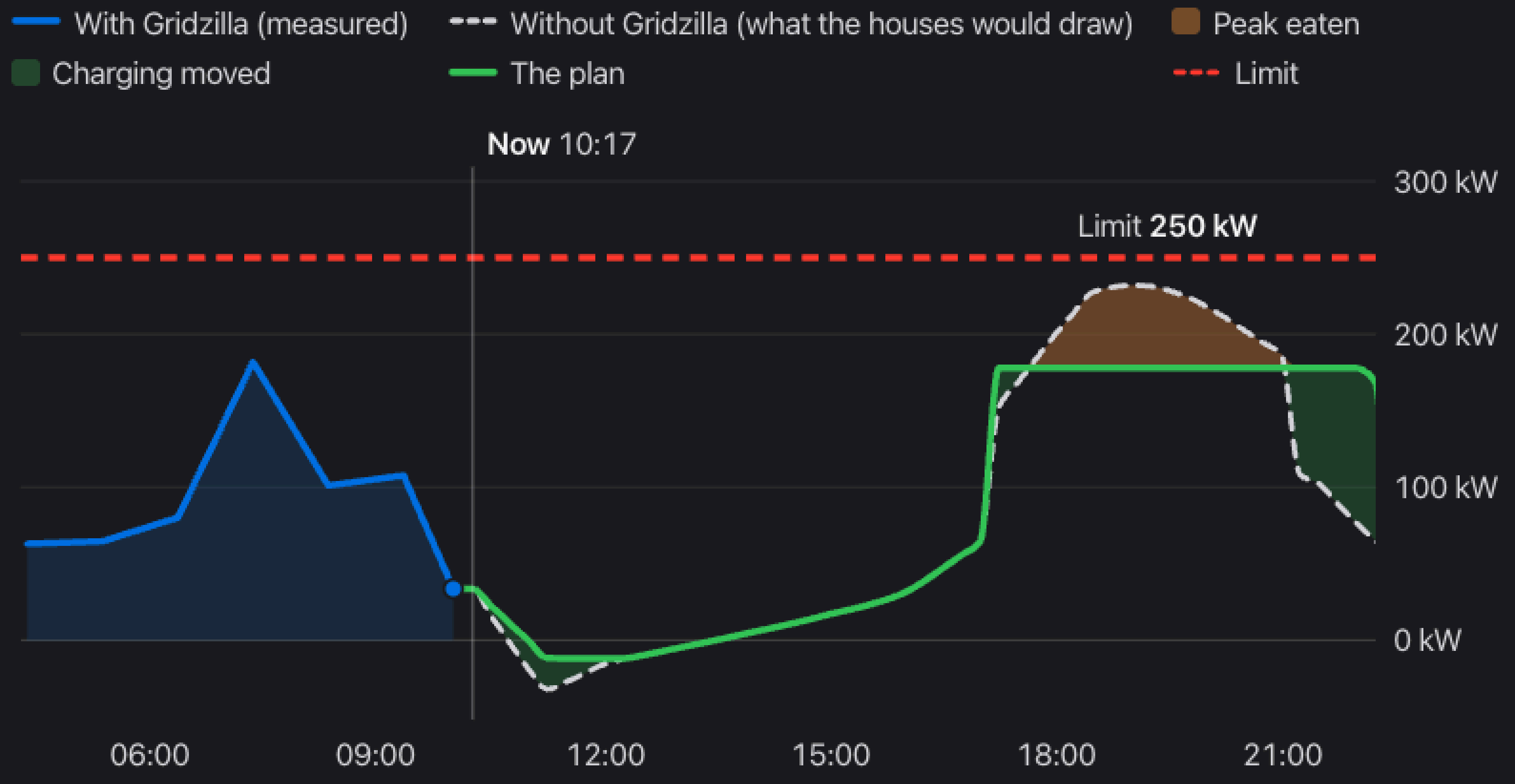
No signal running. Owners run their batteries as they like.

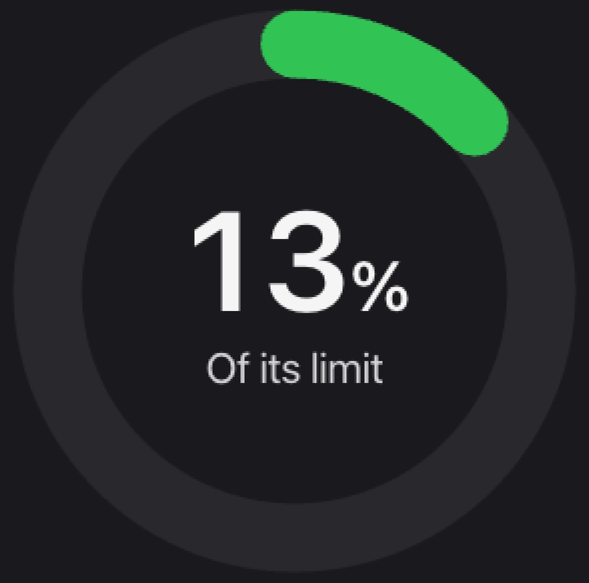
Leobendorf

Stockerau street transformer · limit 250 kW

Peak coming at 18:56: 93 % of the limit. Gridzilla is moving 54 kW to keep Leobendorf at 71 %.

Weather Wien-Liesing: 7.3 °C, feels 3.8 °C · cloud 100 % · sun 196 W/m² · wind 9.8 km/h · no official warnings · 7 °C at 10:00 (feels 4 °C), so heat pumps run hard; some sun, up to 187 W/m² at 11:00.





Blatnica · real plant · live

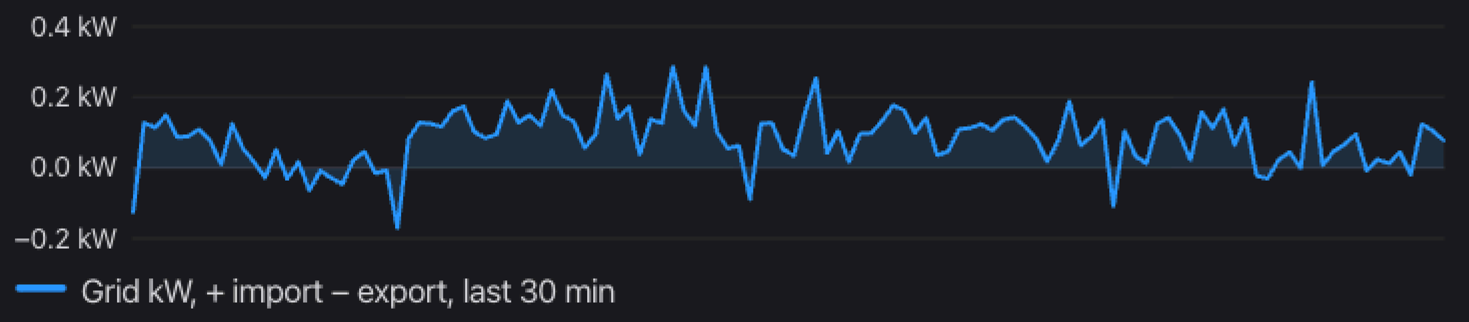
● live · updated 7 s ago · Cerbo GX + Vitocal via Gridzilla agents

+0.08 kW
grid import

35 %
battery SoC · 15 kWh · owner floor 20 %

0.0 kW
EV charger · 11 kW station · idle

0.0 kW
heat pump · tank 50.6 °C → 40 °C · room → 21 °C · floor flow 32 °C



owner mode · last ack reverted (window_end) · signature ✓

What just happened All events

06:21 Diet plan renewed for 24 feeders · 2,168 houses on the plan · a new 24-hour plan per street
04:51 Use your battery on Simmering 28 ended: its time window ran out
03:13 Use your battery on feeder Donaustadt 4 ended: its time window ran out

Wien Sued ●●●●●●●●● Wien Nord ●●●●●●●●● Wien West ●●●●●●●●● Wien Ost ●●●●●●●●● Kendlerstrasse ●●●●●●●●● Wien Mitte ●●●●●●●●● Bisamberg ●●●●●●●●●
Schwarzenbergplatz ●●●●●●●●●

Our family home in Blatnica

Victron battery	Home at the meter
35%	0.1kW
Car charger	**Viessmann heat pump**
0.0kW	0.0kW

✓ Live, on the same signed message path

Signed
by the platform,
one message

Checked
by the plant,
signature first

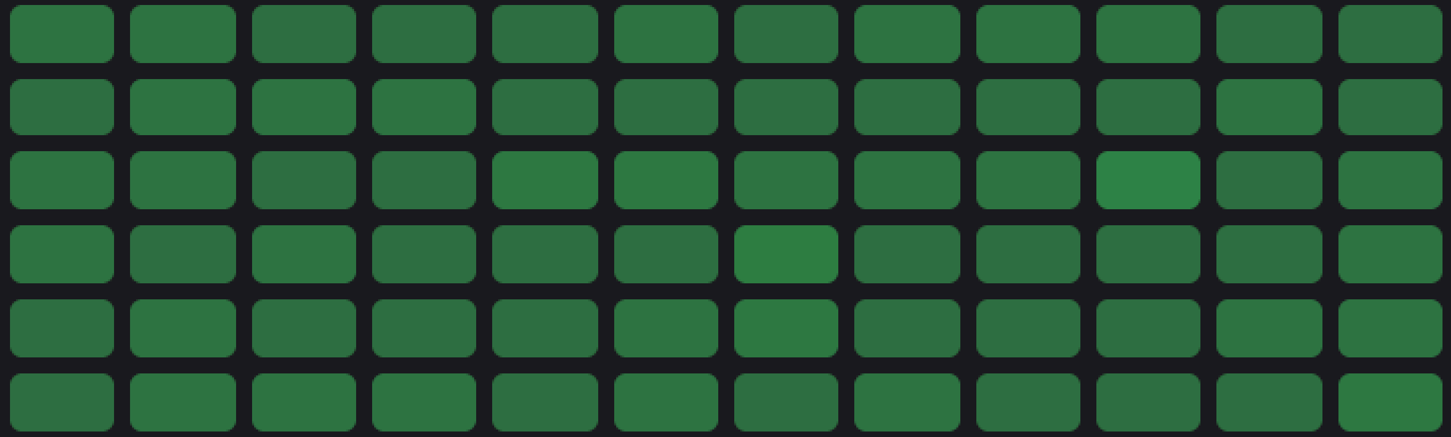
Followed
by the battery
and heat pump

Real end to end. The planning runs on the fleet.

Paid by the meter: a flat yearly fee, crisis prices only when the meter proves the limit was kept.

A simulated Vienna

Two thousand houses, the same cars with and without Gridzilla, realistic charging.



72 street transformers in a simulated Vienna, live

🟩 72 inside the limit 🟧 0 near it 🟥 0 over

-36%
one street
Atzgersdorf,
511 to 325 kW

-5MW
Vienna
two thousand houses,
31.2 to 26.2 MW

0kW
a street
where no car
charged

Measured on the simulated fleet, 24 Oct 2026, with the heat pumps shedding floor heating as well

It eats peaks for breakfast.

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